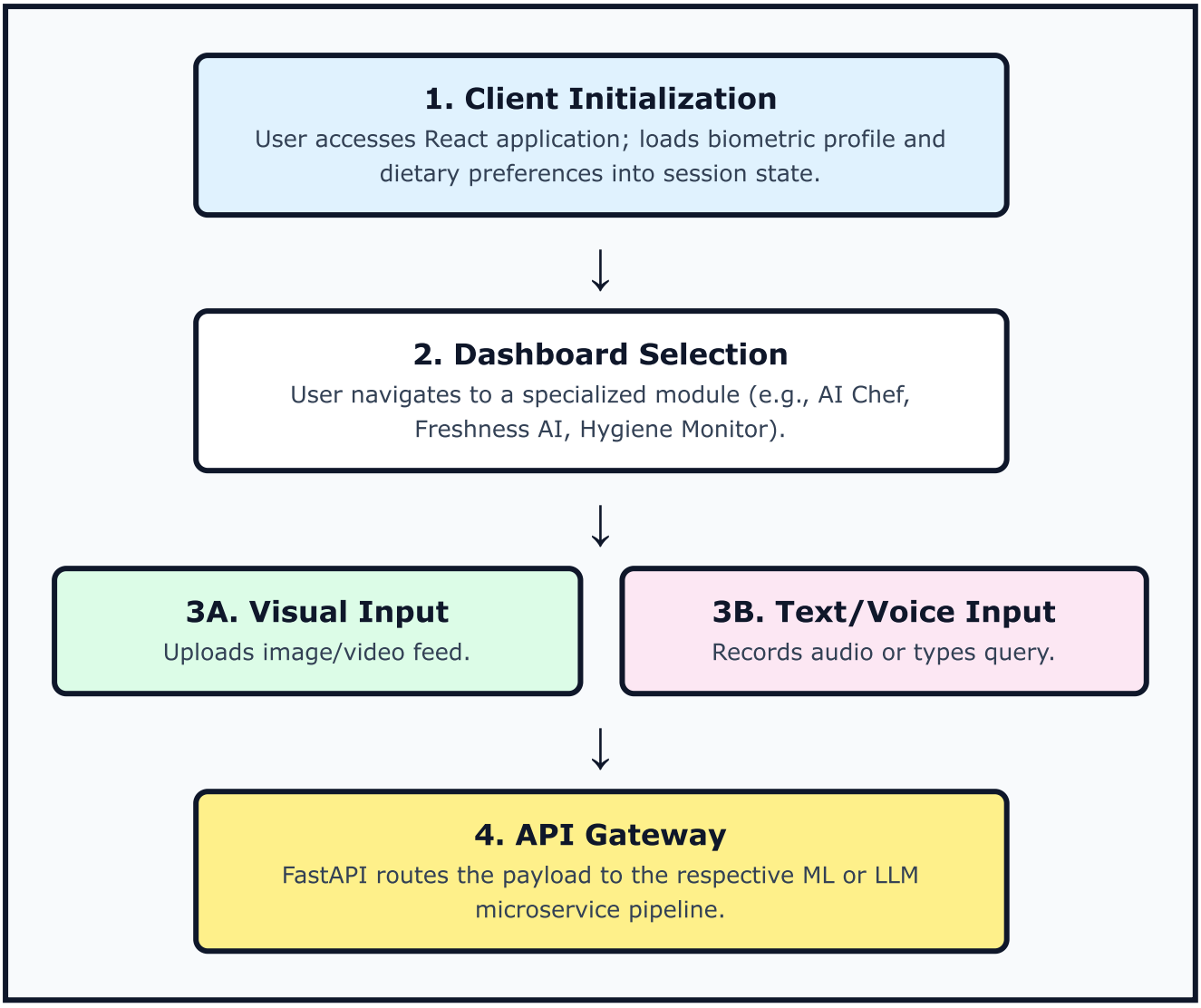
# PROJECT WORKFLOW

The FoodVerse AI workflow is designed around a decoupled, event-driven architecture. User actions on the frontend trigger specific API pipelines on the backend, routing data either to the PyTorch vision models or the Groq-powered LLM engine depending on the active module.

## Phase 1: High-Level Application Routing



# Phase 2: Vision Processing Pipeline

When a user uploads an image for Food Recognition, Freshness Detection, or Hygiene Monitoring, the data strictly follows the visual processing lifecycle to minimize latency and ensure privacy.

### 1. Image Ingestion (FastAPI)

Multipart-form data is received. The API validates file type, limits payload size (Max 12MB), and stores the file temporarily in memory.

↓

### 2. OpenCV Preprocessing

The image is resized to standard dimensions (e.g., 224x224), normalized, and converted into a tensor format suitable for the neural network.

↓

### 3. PyTorch Model Inference

The tensor is passed through the respective trained vision model (e.g., YOLO for hygiene bounding boxes, or ResNet for freshness classification).

↓

### 4. Post-Processing & Formatting

Raw model logits are converted into confidence scores and mapped to human-readable labels (e.g., "Spoiled Apple: 92% Confidence").

↓

### 5. Frontend State Update

FastAPI returns a structured JSON response. React parses the JSON and immediately updates the UI dashboard with the metrics.

## Phase 3: Generative AI Workflow

For cognitive tasks like the Smart AI Chef and Personalized Meal Planner, the workflow aggregates contextual data and utilizes prompt engineering before querying the LLM engine.

